Fundamentals Chapter 4

Siddharth K

October 2024

1. Since G contains an element of order m, there exists $g \in G$ such that $g^m = e$. Now, for any d that divides m, let $g' = g^{m/d}$. Then $(g')^d = (g^{m/d})^d = g^m = e$, so $|g'| = d$.

2. For the sake of contradiction, assume that $G \neq \langle g \rangle$ for all $g \in G$ (where $g > 1$). Then there exists $h \in G$ with $\langle g \rangle \cap \langle h \rangle = \{e\}$ and $h > 1$ so that $h \neq g^k$ for any $k \in \mathbb{Z}$.

Now, consider $\langle g, h \rangle \leq G$. There exists a bijection from $\langle g, h \rangle$ to $\langle g \rangle \times \langle h \rangle$. Each $(g_1, h_1) \in \langle g \rangle \times \langle h \rangle$ can be uniquely mapped to $g_1 h_1 \in \langle g, h \rangle$.

Also, each $a \in \langle g, h \rangle$ can be decomposed uniquely as $a = g_1 h_1$ for some $(g_1, h_1) \in \langle g \rangle \times \langle h \rangle$. This is because, if $a = g_1 h_1 = g_2 h_2$ then $g_1 = g_2 h_2 h_1^{-1}$, which implies that $h_2 h_1^{-1} = e$ so $h_1 = h_2$. Similarly, $g_1 = g_2$.

So there is a bijection from $\langle g, h \rangle$ to $\langle g \rangle \times \langle h \rangle$. This means $|\langle g, h \rangle| = |\langle g \rangle \times \langle h \rangle| = |g||h|$. Also, since $\langle g, h \rangle \leq G$, by Lagrange, $|\langle g, h \rangle| \mid p$, so $|g||h| \mid p$, which means either $|g|$ or $|h|$ is 1, which is a contradiction.

Finally, if there is a subgroup $K \leq G$, by Lagrange, $|K| \mid |G| = p$, so $|K| = 1$ or p, which implies that K is trivial. This means $G \cong \mathbb{Z}/p\mathbb{Z}$.

3. The direct product $A \times B$ is the group $\{(a, b) : a \in A, b \in B\}$ under the operation

$$(a_1, b_1) \star (a_2, b_2) = (a_1 \star_1 a_2, b_1 \star_2 b_2).$$

We need to show that it satisfies closure, existence of identity element, existence of inverse, and associativity. For closure, we see that

$$(a_1, b_1) \star (a_2, b_2) = (a_1 \star_1 a_2, b_1 \star_2 b_2) = (a, b) \in A \times B$$

for some $a \in A$ and $b \in B$ because A and B satisfy closure. The identity element is (e_A, e_B) because

$$(a, b) \star (e_A, e_B) = (a \star_1 e_A, b \star_2 e_B) = (a, b).$$

For all $(a, b) \in A \times B$, the inverse is (a^{-1}, b^{-1}) because

$$(a, b) \star (a^{-1}, b^{-1}) = (a \star_1 a^{-1}, b \star b^{-1}) = (e_A, e_B).$$

Finally, it satisfies associativity because

$$\begin{aligned}((a_1, b_1) \star (a_2, b_2)) \star (a_3, b_3) &= (a_1 \star_1 a_2, b_1 \star_2 b_2) \star (a_3), b_3 \\ &= ((a_1 \star_1 a_2) \star_1 a_3, (b_1 \star_2 b_2) \star_2 b_3 \\ &= (a_1 \star_1 (a_2 \star_1 a_3), b_1 \star_2 (b_2 \star_2 b_3) \\ &= (a_1, b_1) \star (a_2 \star_1 a_3, b_2 \star_2 b_3) \\ &= (a_1, b_1) \star ((a_2, b_2) \star (a_3, b_3)).\end{aligned}$$

4. Below are the generators for each group:
$(\mathbb{Z}/2\mathbb{Z})^\times : 1$
$(\mathbb{Z}/3\mathbb{Z})^\times : 2$
$(\mathbb{Z}/5\mathbb{Z})^\times : 2$
$(\mathbb{Z}/7\mathbb{Z})^\times : 3$
$(\mathbb{Z}/11\mathbb{Z})^\times : 7$
$(\mathbb{Z}/8\mathbb{Z})^\times$: none of the elements of $\{1, 2, \ldots, 7\}$ generate all of $(\mathbb{Z}/8\mathbb{Z})^\times$.

5. The group with just one element would be represented by an empty product of groups.

9. To prove the corollary, we use induction on n. If $n = 2$, then $G = A + B$ so $G \cong A \times B$.

Now, assume that there is an isomorphism $f : G \to A_1 \times \cdots \times A_n$ where $G = A_1 + \cdots + A_n$ and $A_i \cap A_j = \{0\}$. We need to show that there's

$$\phi : G + A_{n+1} \to A_1 \times \cdots \times A_{n+1}.$$

Let $B = A_1 \times \cdots \times A_n$. We know that $G + A_{n+1} \cong B + A_{n+1}$, so we need to find an isomorphism $\phi : G + A_{n+1} \to B + A_{n+1}$. Every $h \in G + A_{n+1}$ can be written as $h = g + a$ for $g \in G, a \in A_{n+1}$. So define ϕ as

$$\phi(h) = \phi(g + a) = f(g) + a.$$

Then

$$\begin{aligned}\phi(h_1 + h_2) &= \phi(g_1 + a_1 + g_2 + a_2) \\ &= \phi(g_1 + g_2 + a_1 + a_2) \\ &= f(g_1 + g_2) + a_1 + a_2 \\ &f(g_1) + a_1 + f(g_2) + a_2 \\ &= \phi(h_1) + \phi(h_2),\end{aligned}$$

so $\phi : A_1+\cdots+A_{n+1} \to (A_1\times\cdots\times A_n)+A_{n+1}$ is a homomorphism. Furthermore, since $A_i \cap A_j = \{0\}$,

$$|G| = |(A_1 + \cdots + A_n) + A_{n+1}| = |A_1 + \cdots + A_n||A_{n+1}|$$

$$= |A_1 \times \cdots \times A_n||A_{n+1}| = |A_1 \times \cdots \times A_{n+1}|.$$

So $G \cong (A_1 \times \cdots \times A_n) + A_{n+1} \cong A_1 \times \cdots \times A_{n+1}$.

10. Assume $\gcd(x_1, \ldots, x_n) \neq 1$. Then there exists some prime p that divides all x_i. Since the only possible prime factors of any x_i are $p_1, \ldots, p_n$, $p = p_k$ for some k. But p_k doesn't divide x_k because x_k's prime factors are all of $p_1, \ldots, p_n$ except p_k, which is a contradiction. So $\gcd(x_1, \ldots, x_n) = 1$.

12a. We see that

$$\phi(a + b) = ((a + b) \bmod n_1, \ldots, (a + b) \bmod n_k)$$

$$= (a \bmod n_1, \ldots, a \bmod n_k) + (b \bmod n_1, \ldots, b \bmod n_k)$$

$$= \phi(a) + \phi(b),$$

so ϕ is a homomorphism.

Also, $|\mathbb{Z}/N\mathbb{Z}| = N$ and

$$|\mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}| = n_1 \cdots n_k = N$$

so ϕ is an isomorphism.

12b. We'll show that there are n consecutive composite numbers for any integer n. Consider the numbers $(n+1)!+2, (n+1)!+3, \ldots, (n+1)!+(n+1)$. There are n of these and they can all be factored, since $(n + 1)!$ is divisible by all numbers from 2 to $n + 1$. So there are n consecutive composite numbers.

12c. The set $(\mathbb{Z}/n\mathbb{Z})^\times$ is the set of all integers between 1 and n that are relatively prime to n.